EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Capacitance of a System

1. A battery is used to charged a parallel-plate capacitor, after which it is disconnected. Then the plates are pulled apart to twice their original separation. This process will double the :

(A) capacitance (B) surface charge density on each plate

(C) stored energy (D) electric field between the two plates

PCC054

2. A capacitor is connected to a battery. The force of attraction between the plates when the separation between them is halved

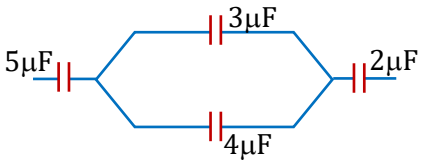
(A) remains the same (B) becomes eight times

(C) becomes four times (D) becomes two times

PCC061

Capacitor Circuits

3. If charge on left plate of the 5 μF capacitor in the circuit segment shown in the figure is –20 μC, the charge on the right plate of 3 μF capacitor is :-

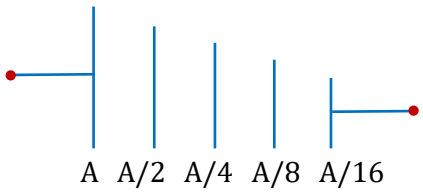


(A) +8.57 μC (B) –8.57 μC (C) +11.42 μC (D) –11.42 μC

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PCC063

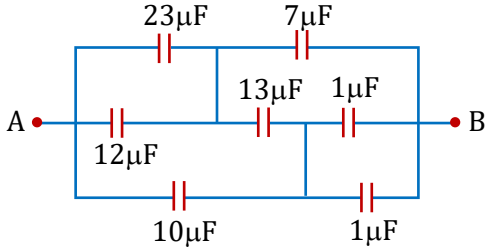
4. 5 Conducting plates each are placed face to face and equi-spaced at distance d. Area of each plate is half the previous plate. If area of first plate is A. Then the equivalent capacitance of the system shown is :-



(A) $\frac{\varepsilon_0 A}{d}$ (B) $\frac{\varepsilon_0 A}{10d}$ (C) $\frac{\varepsilon_0 A}{20d}$ (D) $\frac{\varepsilon_0 A}{30d}$

PCC064

5. Find the equivalent capacitance across A and B :-



(A) $\frac{28}{3}\mu f$ (B) $\frac{15}{2}\mu F$ (C) 15 μF (D) None

PCC067

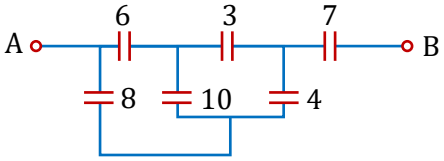
For More Join: @IITJEE_Advanced

6. Three long concentric conducting cylindrical shells have radii R, $2R$ and $2\sqrt{2}R$. Inner and outer shells are connected to each other. The capacitance across middle and inner shell per unit length is:

(A) $\frac{\frac{1}{3}\in_0}{\ln 2}$ (B) $\frac{6\pi\in_0}{\ln 2}$ (C) $\frac{\pi\in_0}{2\ln 2}$ (D) None

PCC068

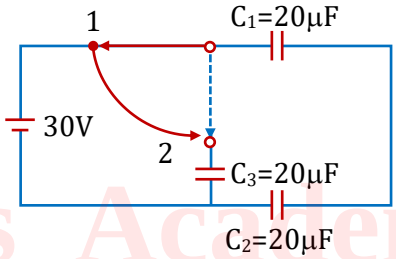
7. In the circuit diagram shown all the capacitors are in μF. The equivalent capacitance between points A and B is (in μF)



(A) 14/5 (B) 7.5 (C) 3/7 (D) None of these

PCC069

8. In the circuit shown in the figure, the switch can be shifted to positions '1' and '2'. Initially the switch was at position '1' for long time and Now, the switch is shifted to position '2' The charge appearing on capacitor C_3 is

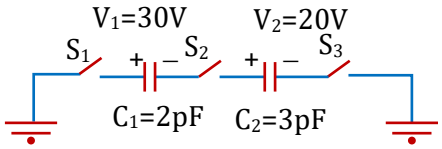


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(A) 225 μC (B) 135 μC (C) 270 μC (D) 75 μC

PCC073

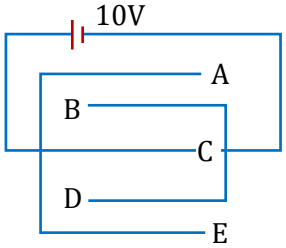
9. For the circuit shown, which of the following statements is true?



(A) with S_1 closed, $V_1 = 15\ V$, $V_2 = 20\ V$
(B) with S_3 closed, $V_1 = V_2 = 25\ V$
(C) with S_1 and S_2 closed, $V_1 = V_2 = 0$
(D) with S_1 and S_2 closed, $V_1 = 30\ V$, $V_2 = 20\ V$

PCC076

10. Five identical capacitor plates are arranged such that they make capacitors each of 2 μF. The plates are connected to a source of emf 10 V. The charge on plate C is



(A) +20 μC (B) +40 μC (C) +60 μC (D) +80 μC

PCC086

For More Join: @IITJEE_Advanced

Energy Analysis in Capacitors

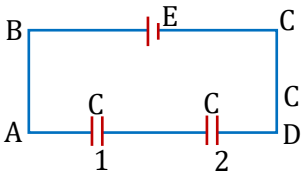
11. A capacitor of capacitance C is initially charged to a potential difference of V volt. Now it is connected to a battery of $2V$ with opposite polarity. The ratio of heat generated to the final energy stored in the capacitor will be

(A) 1.75 (B) 2.25 (C) 2.5 (D) 1/2

PCC088

Dielectrics

12. In the adjoining figure, capacitor (1) and (2) have a capacitance *'C'* each. When the dielectric of dielectric constant K is inserted between the plates of one of the capacitor, the total charge flowing through battery is :-



(A) $\frac{KCE}{K+1}$ from B to C (B) $\frac{KCE}{K+1}$ from C to B

(C) $\frac{(K-1)CE}{2(K+1)}$ from B to C (D) $\frac{(K-1)CE}{2(K+1)}$ from C to B

PCC071

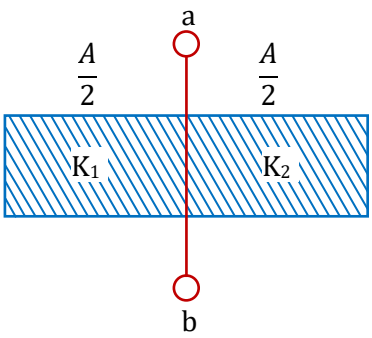
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13. Condenser A has a capacity of 15 when it is filled with a medium of dielectric constant 15. Another condenser B has a capacity 1 with air between the plates. Both are charged separately by a battery of $100V$. After charging, both are connected in parallel without the battery and the dielectric material being removed. The common potential now is :-

(A) 400 V (B) 800 V (C) 1200 V (D) 1600 V

PCC085

14. The capacity of a parallel plate air capacitor is 10 μF. As shown in the figure this capacitor is divided into two equal parts, these parts are filled by media of dielectric constants $K_1 = 2$ and $K_2 = 4$, capacity of this arrangement will be :–

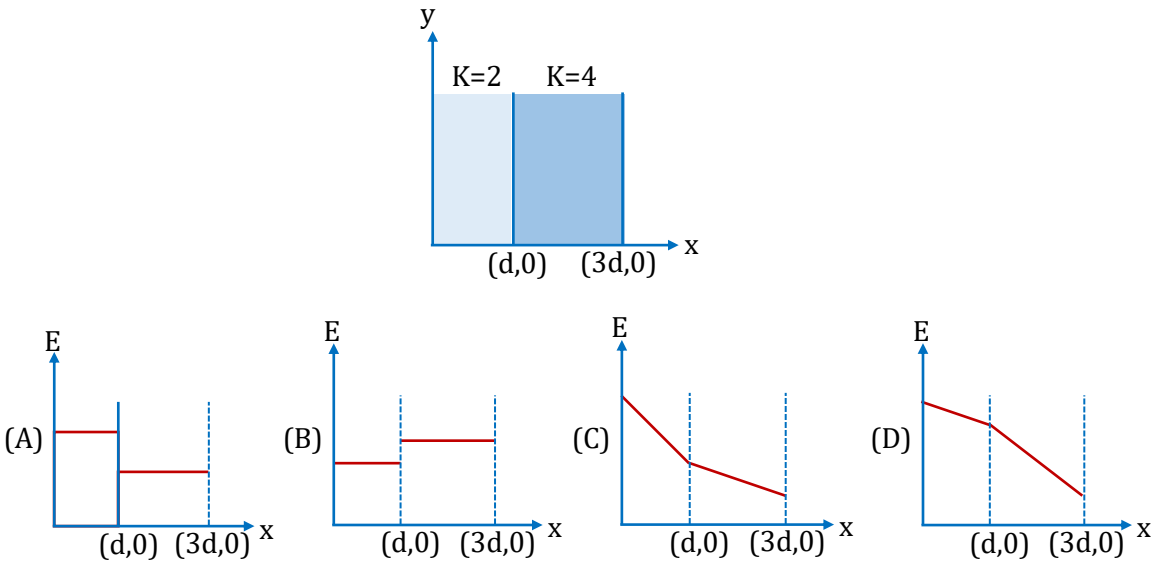


(A) 20 μF (B) 30 μF (C) 10 μF (D) 40 μF

PCC091

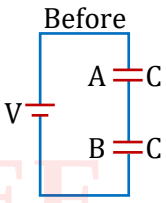
For More Join: @IITJEE_Advanced

15. A parallel plate capacitor has two layers of dielectric as shown in figure. This capacitor is connected across a battery. The graph which shows the variation of electric field (E) and distance (x) from left plate.



PCC092

16. Two identical capacitors are connected in series as shown in the figure. A dielectric slab ($K > 1$) is placed between the plates of the capacitor B and the battery remains connected. Select correct statement.

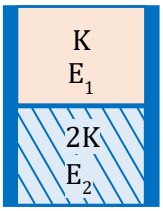


(A) The charge supplied by the battery increases.
(B) The capacitance of the system decreases.
(C) The electric field in the capacitor B increases.
(D) The electrostatic potential energy decreases.

Parikh's Academy JEE

PCC093

17. A parallel plate capacitor is connected from a cell and then isolated from it. Two dielectric slabs of dielectric constant K and $2K$ are now introduce in the region between upper half and lower half of the plate (as shown in figure). The electric field intensity in upper half of dielectric is E_1 and lower half is E_2 then

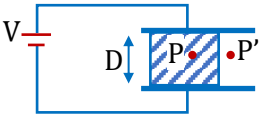


(A) $E_1 = 2\,E_2$
(B) Electrostatic potential energy of upper half is less than that of lower half
(C) Induced charges on both slabs are same
(D) Charge distribution on the plates remains same after insertion of dielectric

PCC096

18. A parallel plate capacitor, partially filled with a dielectric slab of dielectric constant K, is connected with a cell of emf V volt, as shown in the figure. Separation between the plates is D. Then

(A) electric field at point P is less than that at point P'.
(B) electric field at point P' is less than that at point P.
(C) electric fields at points P and P' are equal.
(D) electric field at point P is $E = \frac{V}{KD}$.



PCC097

For More Join: @IITJEE_Advanced

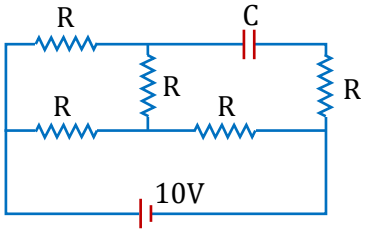
R-C Circuits

19. An uncharged capacitor of capacitance 4 μF, a battery of emf 12 volt and a resistor of 2.5 $M\Omega$ are connected in series. The time after which $v_c = 3v_R$ is (Take $ln2 = 0.693$)

(A) 6.93 sec. (B) 13.86 sec. (C) 20.52 sec. (D) None of these

PCC100

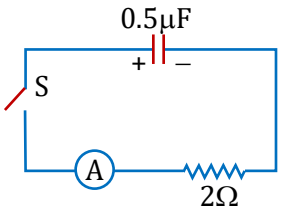
20. Find the potential difference across the capacitor in volts.



(A) $4V$ (B) $2V$ (C) $1V$ (D) $8V$

PCC103

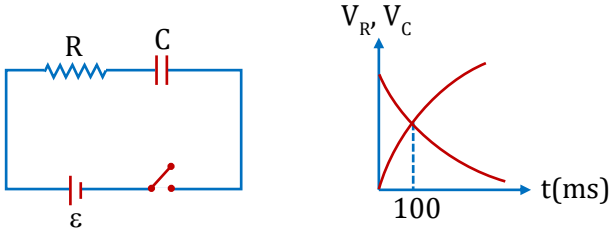
21. A charged capacitor is allowed to discharge through a resistance 2Ω by closing the switch S at the instant $t = 0$. At time $t = ln2\ \mu s$, the reading of the ammeter falls half of its initial value. The resistance of the ammeter equal to



(A) 0 (B) $2\ \Omega$ (C) ∞ (D) $2\ M\Omega$

PCC105

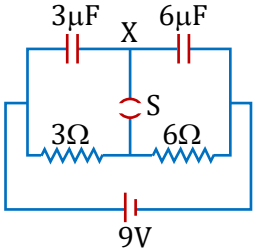
22. In the RC circuit shown, switch is closed at $t = 0$. Graphs showing variation of potential (V_R) across resistor and potential (V_C) across capacitor are given. Time constant of circuit is approximately equal to :-



(A) 100 ms (B) 145 ms (C) 200 ms (D) 300 ms

PCC108

23. A circuit is connected as shown in the figure with the switch S open. When the switch is closed, the total amount of charge that flows from Y to X is :-



(A) 0 (B) 54 μC (C) 27 μC (D) 81 μC

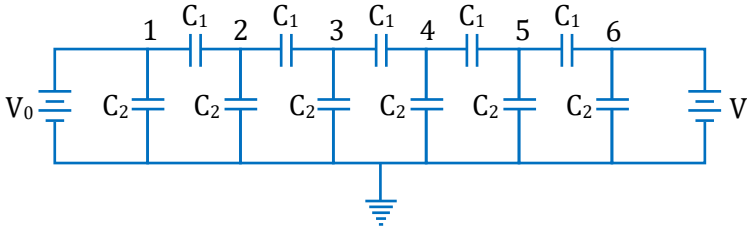
PCC117

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. In the circuit shown, ratio of capacitors is $\frac{C_2}{C_1}=\frac{4}{3}$ and terminal voltage of the battery connected at the left end is V_0. Given that the potentials of nodes 1, 2, 3, 4, 5 and 6 are in geometric progression, when the terminal voltage of the battery connected at the right most end is V. (In steady state condition) Then :-



(A) V may be $\frac{V_0}{3^5}$

(B) V may be $V_0 3^5$

(C) V may be $\frac{V_0}{4}$

(D) V cannot be concluded with this data

PCC126

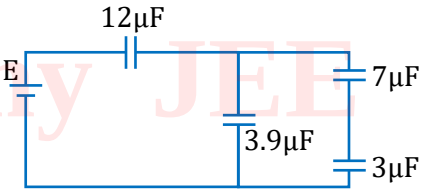
2. Four capacitors and a battery are connected as shown. The potential drop across the $7\mu F$ capacitor is $6V$. Then the

(A) Potential difference across the $3\mu F$ capacitor is $10V$

(B) Charge on the $3\mu F$ capacitor is $42\mu C$

(C) e.m.f of the battery is $30V$

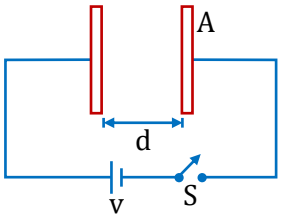
(D) Potential difference across the $12\mu F$ capacitor is $10V$



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PCC127

3. A parallel plate air capacitor has plate area A and separation between the plates d. Switch S is closed to connect the capacitor to a cell of emf V.



(A) Amount of heat generated in the circuit as the capacitor gets charged is $\frac{1}{2}CV^2$.

(B) Magnitude of force that one capacitor plate exerts on the other is $\frac{\varepsilon_0 AV^2}{d^2}$.

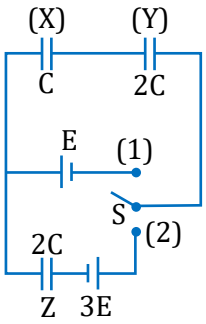
(C) If the distance between the plates is slowly reduced to $\frac{d}{2}$, the work done by external agent in the process is given by $-\frac{1}{2}CV^2$.

(D) If the distance between the plates is reduced slowly from d to $\frac{d}{2}$, energy dissipated in the circuit is zero.

PCC128

For More Join: @IITJEE_Advanced

4. In the given circuit diagram initially all the capacitors are uncharged. Now switch S is closed with position (1) and when steady state reached switch is shifted to position (2) from (1).



(A) Final charge on all three capacitors will be same.

(B) Final charge on capacitor Y is $\frac{4CE}{3}$.

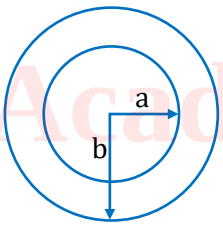
(C) Total heat developed in the circuit after shifting the switch from position (1) to position (2) is $4CE^2$.

(D) Total heat developed in circuit from starting is $\frac{26}{3}CE^2$.

PCC130

5. In the system of concentric conducting spheres, the :-

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(A) Capacitance of the system is $\frac{4\pi\varepsilon_0 ab}{b-a}$ when the outer sphere is earthed

(B) The energy stored in the system if a charge Q is given to the inner sphere is $\frac{Q^2}{8\pi\varepsilon_0}\left(\frac{1}{a}-\frac{1}{b}\right)$ in (a)

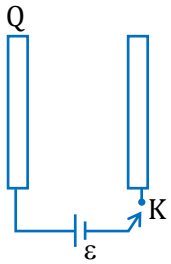
(C) If the inner sphere is earthed, equal and opposite charges will be held by the conductors

(D) None of these

PCC131

6. In the given figure the capacitance of capacitor is C. The left plate of the capacitor is given charge Q and right plate is uncharged. Then choose the correct statement(s) :-

(A) The amount of charge that will flow through the battery till the steady state is reached when the switch K is closed will be $C\varepsilon + \frac{Q}{2}$



(B) The charge appearing on the inner face of the left plate at steady state is $C\varepsilon$

(C) The charge appearing on the interface of the right plate at steady state is $C\varepsilon$

(D) If $\varepsilon = \frac{2Q}{C}$, then work done by cell after closing the switch is $\frac{3Q^2}{C}$

PCC132

For More Join: @IITJEE_Advanced

7. The capacitance of a parallel plate capacitor is C when the region between the plate has air. This region is now filled with a dielectric slab of dielectric constant k. The capacitor is connected to a cell of *emf* E, and the slab is taken out:

(A) charge $CE(k-1)$ flows through the cell.

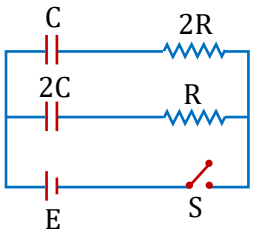
(B) energy $E^2C(k-1)$ is absorbed by the cell.

(C) the energy stored in the capacitor is reduced by $E^2C(k-1)$.

(D) the external agent has to do $\frac{1}{2}E^2C(k-1)$ amount of work to take the slab out.

PCC058

8. In the circuit shown in the figure switch S is closed at time $t = 0$. Select the correct statement :-



(A) Rate of increase of charge is same in both the capacitor.

(B) Ratio of charge stored in capacitance C and $2C$ at any time t would be 1 : 2.

(C) Time constant of both the capacitors are equal.

(D) Steady state charge in capacitors C and $2C$ are in the ratio 1 : 2.

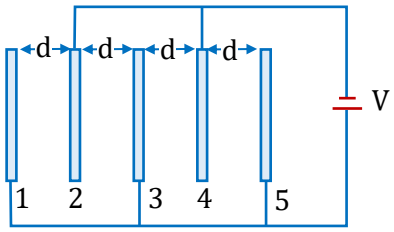
PCC109

COMPREHENSION TYPE QUESTIONS

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PARAGRAPH FOR QUESTION 9 TO 11

Five identical capacitor plates, each of area A, are arranged such that adjacent plates are at d distance apart. Plates are connected to a source of emf V as shown in figure.



9. The charge on plate 1 is

(A) $\frac{\epsilon_0 AV}{d}$ (B) $\frac{2\epsilon_0 AV}{d}$ (C) $-\frac{\epsilon_0 AV}{d}$ (D) $-\frac{2\epsilon_0 AV}{d}$

PCC082

10. The charge on plate 4 is

(A) $\frac{\epsilon_0 AV}{d}$ (B) $\frac{2\epsilon_0 AV}{d}$ (C) $-\frac{\epsilon_0 AV}{d}$ (D) $-\frac{2\epsilon_0 AV}{d}$

PCC083

11. Total electrostatic energy stored in the system is

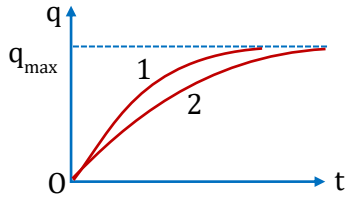
(A) $\frac{A\epsilon_0 V^2}{2d}$ (B) $\frac{2A\epsilon_0 V^2}{d}$ (C) $\frac{A\epsilon_0 V^2}{10d}$ (D) $\frac{3A\epsilon_0 V^2}{2d}$

PCC084

For More Join: @IITJEE_Advanced

PARAGRAPH FOR QUESTION NO. 12 AND 13

The charge across the capacitor in two different *RC* circuits 1 and 2 are plotted as shown in figure.



12. Choose the correct statement(*s*) related to the two circuits.

(A) Both the capacitors are charged to the same charge.

(B) The emf's of cells in both the circuit are equal.

(C) The emf's of the cells may be different.

(D) The emf E_1 is more than E_2.

PCC113

13. Identify the correct statement(*s*) related to the R_1, R_2, C_1 and C_2 of the two *RC* circuits.

(A) $R_1 > R_2$ if $E_1 = E_2$

(B) $C_1 < C_2$ if $E_1 = E_2$

(C) $R_1C_1 > R_2C_2$

(D) $\frac{R_1}{R_2}3 < \frac{C_2}{C_1}$

PCC114

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For More Join: @IITJEE_Advanced

EXERCISE (S)

Capacitance of a System

1. Eight drops of mercury of equal radii possessing equal charges combine to form a big drop. Then the capacitance of bigger drop compared to each individual small drop is :- (in times)

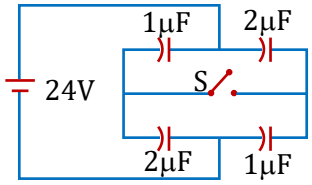
PCC133

2. 2 conducting objects one with charge of $+Q$ and another with $-Q$ are kept on x-axis at $x = 0$ and $x = 1$ respectively. The electric field on the x-axis is given by $3Q\left(x^2+\frac{4}{3}\right)$. If the capacitance of this configuration of objects is C. Value of $\frac{1}{C}$ is

PCC134

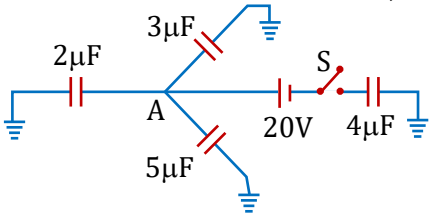
Capacitor Circuits

3. The connections shown in figure are established with the switch S open. How much charge will flow through the switch if it is closed?



PCC070

4. Three capacitors of 2 μF, 3 μF and 5 μF are independently charged with batteries of emf's 5V, 20V and 10V respectively. After disconnecting from the voltage sources. These capacitors are connected as shown in figure with their positive polarity plates are connected to A and negative polarity is earthed. Now a battery of 20V and an uncharged capacitor of 4 μF capacitance are connected to the junction A as shown with a switch S. When switch is closed, find :



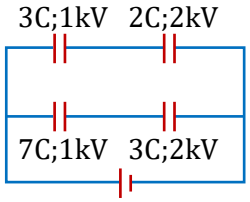
(a) the potential of the junction A.
(b) final charges on all four capacitors.

PCC077

5. A 10 μF capacitor is fully charged across a 12-volt battery. The capacitor is then disconnected from the battery and connected across an initially uncharged capacitor, C. The voltage across each capacitor is now 3 volts. What is the unknown capacitance C (in μF)?

PCC079

6. The diagram shows four capacitors with capacitances and break down voltages as mentioned. What should be the maximum value of the external emf source such that no capacitor breaks down?



PCC080

For More Join: @IITJEE_Advanced

Energy Analysis in Capacitors

7. Inside a parallel plate capacitor with an area of 200 cm^2 and the distance between them 0.885 mm is a glass plate, completely filling the space between the plates of the capacitor. What minimum work (in μJ) is required to remove the glass plate? Capacitor is permanently connected to the battery with electromotive force 200 V. The dielectric constant of glass is 5. Force of friction and the mass of the glass plate is neglected.

PCC135

8. A 600 pF capacitor is charged by 200 V supply. It is then disconnected from the supply and is connected to another uncharged 600 pF capacitor. Electrostatic energy lost in the process is _____ μJ.

PCC136

Dielectrics

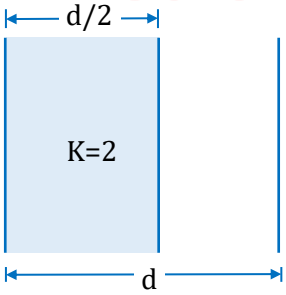
9. A parallel plate capacitor is to be designed which is to be connected across 1 kV potential difference. The dielectric material which is to be filled between the plates has dielectric constant $K = 6\pi$ and dielectric strength 10^7 V/m. For safety the electric field is never to exceed 10% of the dielectric strength. With such specifications, if we want a capacitor of capacitance 50 pF, what minimum area (in mm^2) of plates is required for safe working?

(Use $\varepsilon_0 = \frac{1}{36\pi} \times 10^{-9}$ in S.I. unit)

PCC056

10. A certain series RC circuit is formed using a resistance R, a capacitor without dielectric having a capacitance $C = 2$ F and a battery of emf $E = 3V$. The circuit is completed and it is allowed to attain the steady state. After this, at $t = 0$, half the thickness of the capacitor is filled with a dielectric of constant $K = 2$ heat generated (in joule) in the circuit between $t = 0$ and $t = \infty$?

Pankers Academy JEE

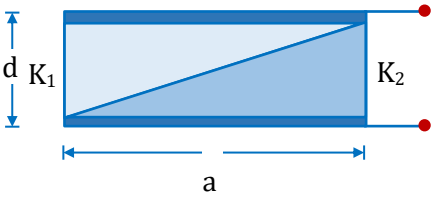


PCC087

11. Two parallel plate air capacitors each of capacitance C were connected in series to a battery with e.m.f ε. Then one of the capacitors was filled up with uniform dielectric with relative permittivity k. What amount of charge flows through the battery ?

PCC098

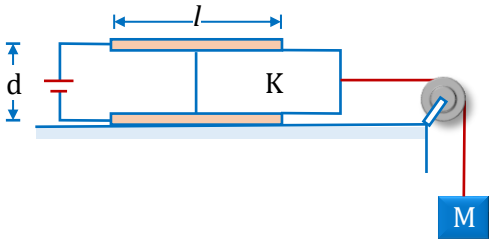
12. A capacitor is formed by two square metal-plates of edge a, separated by a distance d. Dielectrics of dielectric constants K_1 and K_2 are filled in the gap as shown in figure. Find the capacitance.



PCC099

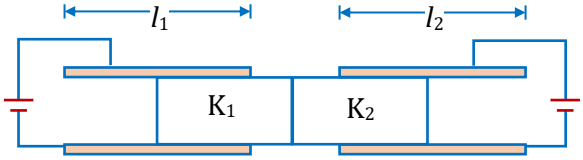
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13. Consider the situation shown in figure. The width of each plate is b. The capacitor plates are rigidly clamped in the laboratory and connected to a battery of emf E. All surfaces are frictionless. Calculate the value of M for which the dielectric slab will stay in equilibrium.



PCC120

14. Figure shows two parallel plate capacitors with fixed plates and connected to two batteries. The separation between the plates is the same for the two capacitors. The plates are rectangular in shape with width b and lengths l_1 and l_2. The left half of the dielectric slab has a dielectric constant K_1 and the right half K_2. Neglecting any friction, find the ratio of the emf of the left battery to that of the right battery for which the dielectric slab may remain in equilibrium.



PCC121

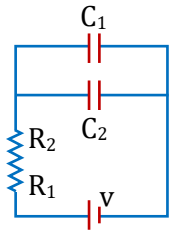
R-C Circuits

15. A capacitor of capacitance 12.0 μF is connected to a battery of emf 6.00 V and internal resistance 1.00 Ω through resistance less leads. 12.0 μs after the connections are made, what will be (a) the rate at which the energy stored in the capacitor increasing.

Bansal's Academy JEE

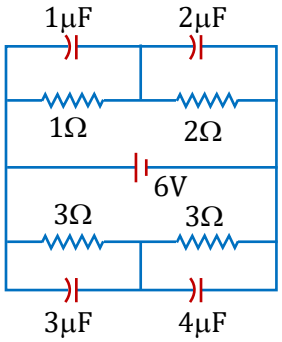
PCC101

16. Given $R_1 = 1\ \Omega$, $R_2 = 2\ \Omega$, $C_1 = 2\ \mu F$, $C_2 = 4\ \mu F$. The time constants (in μS) for the circuits is :-



PCC106

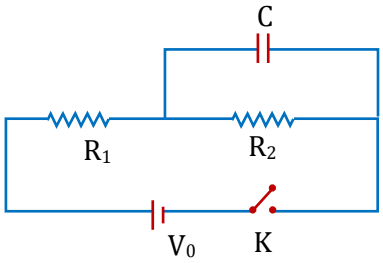
17. Find the charges on the four capacitors of capacitances 1μF, 2μF, 3μF and 4μF shown in figure.



PCC111

For More Join: @IITJEE_Advanced

18. In the connection shown in the figure the switch K is open and the capacitor is uncharged. Then we close the switch and let the capacitor charge up to the maximum and open the switch again. Then (Use the following data : $V_0 = 30\ V$, $R_1 = 10\ k\Omega$, $R_2 = 5\ k\Omega$)



(i) the current through R_1 be I_1 immediately after closing the switch

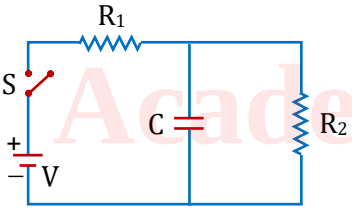
(ii) the current through R_2 be I_2 a long time after the switch was closed

(iii) the current through R_2 be I_3 immediately after reopening the switch

Find the value of $\frac{I_1}{I_2 I_3}$ (in ampere–1).

PCC115

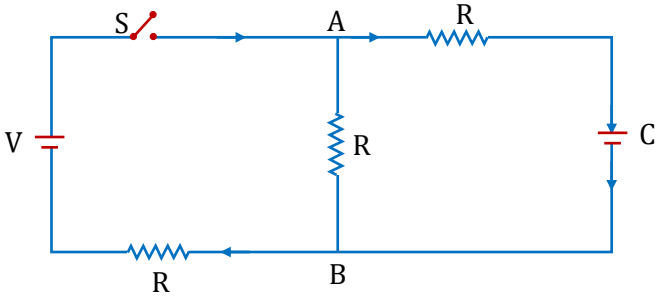
19. In the given circuit, the switch S is closed at time $t = 0$. The charge Q on the capacitor at any instant t is given by $Q(t) = Q_0(1 - e^{-\alpha t})$. Find the value of Q_0 and α in terms of given parameters shown in the circuit.



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PCC118

20. In the circuit shown in figure, the battery is an ideal one with emf V. The capacitor is initially uncharged. Switch S is closed at time $t = 0$



(a) Find the charge Q on the capacitor at time t.

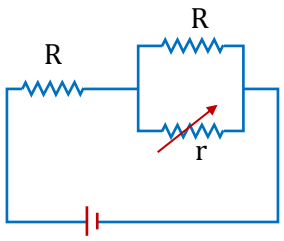
(b) Find the current in AB at time t. What is its limiting value as $t \to$ infinity?

PCC119

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. In the circuit shown, the resistance r is a variable resistance. If for $r = fR$, the heat generation in r is maximum then the value of f is : **[JEE (Main) 2016]**



(1) 1 (2) $\frac{1}{4}$ (3) $\frac{1}{2}$ (4) $\frac{3}{4}$

PCC021

2. Three capacitors each of 4 μF are to be connected in such a way that the effective capacitance is 6 μF. This can be done by connecting them : **[JEE (Main) 2016]**

(1) two in parallel and one in series (2) all in parallel

(3) two in series and one in parallel (4) all in series

PCC022

3. A capacitance of 2 μF is required in an electrical circuit across a potential difference of 1.0 kV. A large number of 1 μF capacitors are available which can withstand a potential difference of not more than 300 V. The minimum number of capacitors required to achieve this is : **[JEE (Main) 2017]**

(1) 24 (2) 32 (3) 2 (4) 16

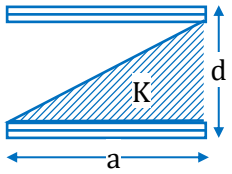
PCC023

4. A parallel plate capacitor of capacitance 90 pF is connected to a battery of emf 20V. If a dielectric material of dielectric constant $K = \frac{5}{3}$ is inserted between the plates, the magnitude of the induced charge will be:- **[JEE (Main) 2018]**

(1) 0.3 $n\,C$ (2) 2.4 $n\,C$ (3) 0.9 $n\,C$ (4) 1.2 $n\,C$

PCC024

5. A parallel plate capacitor is made of two square plates of side 'a', separated by a distance d (d<<a). The lower triangular portion is filled with a dielectric of dielectric constant K, as shown in the figure. Capacitance of this capacitor is : **[JEE (Main) 2019]**

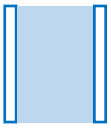


(1) $\frac{1}{2}\frac{k\in_0 a^2}{d}3$ (2) $\frac{k\in_0 a^2}{d}\ln K$ (3) $\frac{k\in_0 a^2}{d(K-1)}\ln K$ (4) $\frac{k\in_0 a^2}{2d(K+1)}$

PCC025

For More Join: @IITJEE_Advanced

6. A parallel plate capacitor has plates of area A separated by distance 'd' between them. It is filled with a dielectric which has a dielectric constant that varies as $k(x) = K(1 + \alpha x)$ where 'x' is the distance measured from one of the plates. If $(\alpha d) << 1$, the total capacitance of the system is best given by the expression : **[JEE (Main) 2020]**



(1) $\frac{AK\varepsilon_0}{d}\left(1+\frac{\alpha d}{2}\right)$ (2) $\frac{A\varepsilon_0 K}{d}\left(1+\left(\frac{\alpha d}{2}\right)^2\right)$ (3) $\frac{A\varepsilon_0 K}{d}\left(1+\frac{\alpha^2 d^2}{2}\right)$ (4) $\frac{AK\varepsilon_0}{d}(1+\alpha d)$

PCC026

7. A parallel plate capacitor is formed by two plates each of area $30\pi\ cm^2$ separated by 1 mm. A material of dielectric strength $3.6 \times 10^7\ Vm^{-1}$ is filled between the plates. If the maximum charge that can be stored on the capacitor without causing any dielectric breakdown is $7 \times 10^{-6}\ C$, the value of dielectric constant of the material is : **[JEE (Main) 2022]**

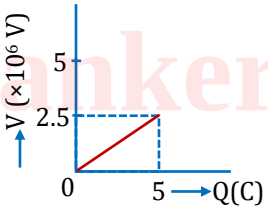
$\left\{\text{Use: } \frac{1}{4\pi\varepsilon_0} = 9\times10^9\ Nm^2C^{-2}\right\}$

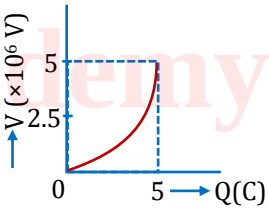
(1) 1.66 (2) 1.75 (3) 2.25 (4) 2.33

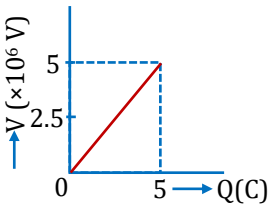
PCC027

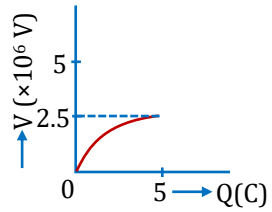
8. A condenser of 2 μF capacitance is charged steadily from 0 to $5C$. Which of the following graph represents correctly the variation of potential difference (V) across it's plates with respect to the charge (Q) on the condenser ? **[JEE (Main) 2022]**

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(1) 

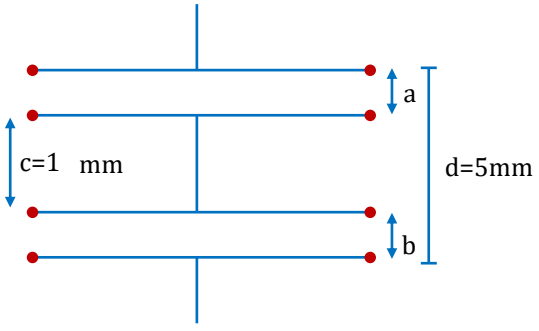
(2) 

(3) 

(4) 

PCC140

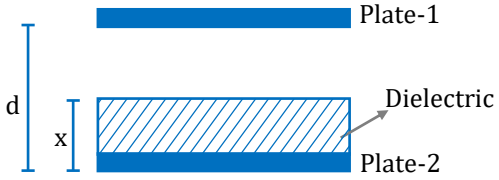
9. As shown in the figure, two parallel plate capacitors having equal plate area of 200 cm^2 are joined in such a way that $a \neq b$. The equivalent capacitance of the combination is $x \in_0 F$. The value of x is_____. **[JEE (Main) 2023]**



PCC122

For More Join: @IITJEE_Advanced

10. A parallel plate capacitor with plate area A and plate separation d is filled with a dielectric material of dielectric constant $K = 4$. The thickness of the dielectric material is x, where $x < d$.



Let C_1 and C_2 be the capacitance of the system for $x = \frac{1}{3}d$ and $x = \frac{2d}{3}$, respectively. If $C_1 = 2\mu F$ the value of C_2 is _____ μF **[JEE (Main) 2023]**

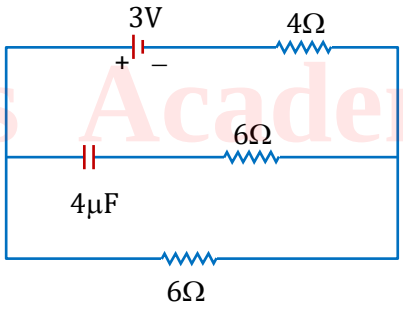
PCC123

11. A parallel plate capacitor has plate area $40 cm^2$ and plates separation 2 mm. The space between the plates is filled with a dielectric medium of a thickness 1 mm and dielectric constant 5. The capacitance of the system is : **[JEE (Main) 2023]**

(1) $24\varepsilon_0 F$ (2) $\frac{3}{10}\varepsilon_0 F$ (3) $\frac{10}{3}\varepsilon_0 F$ (4) $10\varepsilon_0 F$

PCC141

12. In the network shown below, the charge accumulated in the capacitor in steady state will be: **[JEE (Main) 2023]**



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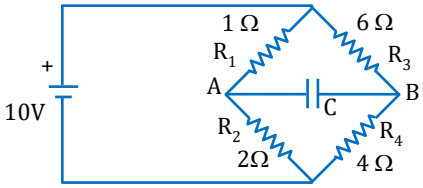
(1) 7.2 μC (2) 4.8 μC (3) 10.3 μC (4) 12 μC

PCC142

13. A parallel plate capacitor with plate separation 5 mm is charged up by a battery. It is found that on introducing a dielectric sheet of thickness 2 mm, while keeping the battery connections intact, the capacitor draws 25% more charge from the battery than before. The dielectric constant of the sheet is ___. **[JEE (Main) 2024]**

PCC143

14. The charge accumulated on the capacitor connected in the following circuit is ___μC (Given $C = 150\ \mu F$) **[JEE (Main) 2024]**

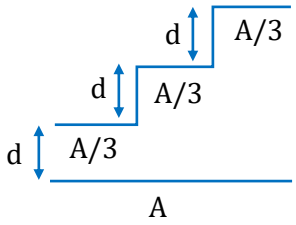


PCC144

For More Join: @IITJEE_Advanced

15. A capacitor is made of a flat plate of area A and a second plate having a stair-like structure as shown in figure. If the area of each stair is $\frac{A}{3}$ and the height is d, the capacitance of the arrangement is:

[JEE (Main) 2024]



(1) $\frac{11\varepsilon_0 A}{18d}$ (2) $\frac{13\varepsilon_0 A}{17d}$

(3) $\frac{11\varepsilon_0 A}{20d}$ (4) $\frac{18\varepsilon_0 A}{11d}$

PCC145

16. A capacitor has air as dielectric medium and two conducting plates of area 12 cm^2 and they are 0.6 cm apart. When a slab of dielectric having area 12 cm^2 and 0.6 cm thickness is inserted between the plates, one of the conducting plates has to be moved by 0.2 cm to keep the capacitance same as in previous case. The dielectric constant of the slab is :

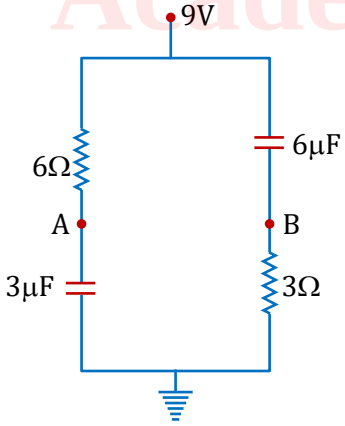
(Given $\in_0 = 8.834 \times 10^{-12}\ F/m$) **[JEE (Main) 2024]**

(1) 1.50 (2) 1.33 (3) 0.66 (4) 1

PCC146

17. In the given figure, the charge stored in $6\mu F$ capacitor, when points A and B are joined by a connecting wire is _____μC. **[JEE (Main) 2024]**

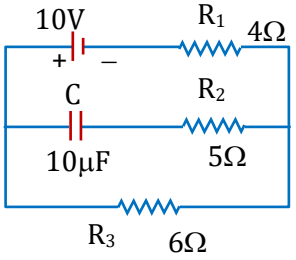
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PCC147

18. In an electrical circuit drawn below the amount of charge stored in the capacitor is __μC.

[JEE (Main) 2024]



PCC148

For More Join: @IITJEE_Advanced

19. Consider a parallel plate capacitor of area A (of each plate) and separation 'd' between the plates. If E is the electric field and ε_0 is the permittivity of free space between the plates, then potential energy stored in the capacitor is :- **[JEE (Main) 2025]**

(1) $\frac{1}{2}\varepsilon_0 E^2 Ad$ (2) $\frac{3}{4}\varepsilon_0 E^2 Ad$ (3) $\frac{1}{4}\varepsilon_0 E^2 Ad$ (4) $\varepsilon_0 E^2 Ad$

PCC179

20. A parallel plate capacitor consisting of two circular plates of radius 10 cm is being charged by a constant current of 0.15 A. If the rate of change of potential difference between the plates is 7×10^8 V/s then the integer value of the distance between the parallel plates is –

$\left(\text{Take}, \in_0 = 9\times10^{-12}\frac{F}{m}, \pi = \frac{22}{7}\right)$ _____ μm. **[JEE (Main) 2025]**

PCC180

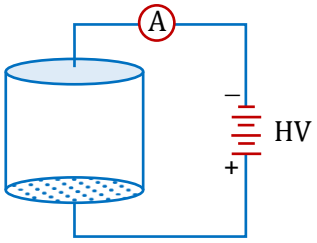
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EXERCISE - JEE (Advanced) PYQ

PARAGRAPH FOR QUESTION NO. 1 AND 2

Consider an evacuated cylindrical chamber of height h having rigid conducting plates at the ends and an insulating curved surface as shown in the figure. A number of spherical balls made of a light weight and soft material and coated with a conducting material are placed on the bottom plate. The balls have a radius $r << h$. Now a high voltage source (HV) is connected across the conducting plates such that the bottom plate is at $+V_0$ and the top plate at $-V_0$. Due to their conducting surface the balls will get charged, will become equipotential with the plate and are repelled by it. The balls will eventually collide with the top plate, where the coefficient of restitution can be taken to be zero due to the soft nature of the material of the balls. The electric field in the chamber can be considered to be that of a parallel plate capacitor. Assume that there are no collisions between the balls and the interaction between them is negligible. (Ignore gravity) **[JEE (Advanced) 2016]**



1. Which one of the following statements is correct?

(A) The balls will execute simple harmonic motion between the two plates

(B) The balls will bounce back to the bottom plate carrying the opposite charge they went up with

(C) The balls will bounce back to the bottom plate carrying the same charge they went up with

(D) The balls will stick to the top plate and remain there

PCC040

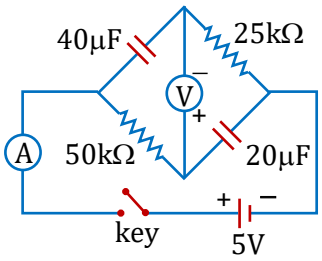
2. The average current in the steady state registered by the ammeter in the circuit will be

(A) proportional to V_0^2 (B) proportional to $V_0^{1/2}$

(C) proportional to the potential V_0 (D) zero

PCC041

3. In the circuit shown below, the key is pressed at time $t = 0$. Which of the following statement(s) is(are) true? **[JEE (Advanced) 2016]**



(A) The voltmeter displays $-5V$ as soon as the key is pressed, and displays $+5V$ after A long time

(B) The voltmeter will display 0 V at time $t = \ln 2$ seconds

(C) The current in the ammeter becomes 1/e of the initial value after 1 second

(D) The current in the ammeter becomes zero after A long time

PCC042

For More Join: @IITJEE_Advanced

PARAGRAPH FOR QUESTION NO. 4 AND 5

Consider *A* simple *RC* circuit as shown in figure 1.

Process 1 : In the circuit the switch *S* is closed at $t = 0$ and the capacitor is fully charged to voltage V_0 (i.e., charging continues for time $T >> RC$). In the process some dissipation (E_D) occurs across the resistance *R*. The amount of energy finally stored in the fully charged capacitor is E_C.

Process 2 : In *A* different process the voltage is first set to $\frac{V_0}{3}$ and maintained for *A* charging time $T >> RC$. Then the voltage is raised to $\frac{2V_0}{3}$ without discharging the capacitor and again maintained for *A* time $T >> RC$. The process is repeated one more time by raising the voltage to V_0 and the capacitor is charged to the same final voltage V_0 as in Process 1.

These two processes are depicted in Figure 2. **[JEE (Advanced) 2017]**

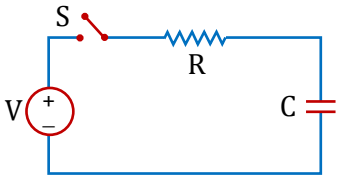


Figure 1

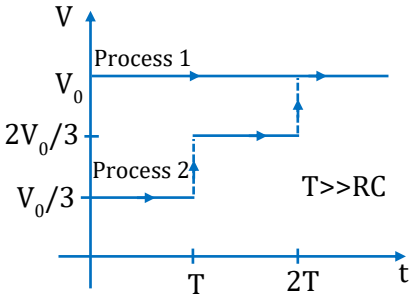


Figure 2

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4. In Process 1, the energy stored in the capacitor E_C and heat dissipated across resistance E_D are related by:-

(A) $E_C = E_D$ (B) $E_C = 2E_D$ (C) $E_C = \frac{1}{2}E_D$ (D) $E_C = E_D \ln 2$

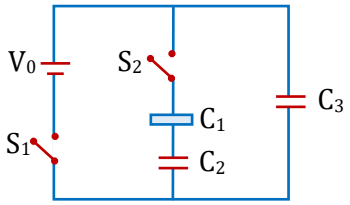
PCC043

5. In Process 2, total energy dissipated across the resistance E_D is :-

(A) $E_D = \frac{1}{3}\left(\frac{1}{2}CV_0^2\right)$ (B) $E_D = 3\left(\frac{1}{2}CV_0^2\right)$ (C) $E_D = \frac{1}{2}CV_0^2$ (D) $E_D = 3CV_0^2$

PCC044

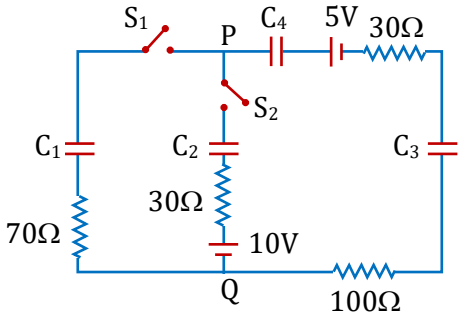
6. Three identical capacitors C_1, C_2 and C_3 have *A* capacitance of 1.0 μF each and they are uncharged initially. They are connected in *A* circuit as shown in the figure and C_1 is then filled completely with *A* dielectric material of relative permittivity $\in_r$. The cell electromotive force (emf) $V_0 = 8V$. First the switch S_1 is closed while the switch S_2 is kept open. When the capacitor C_3 is fully charged, S_1 is opened and S_2 is closed simultaneously. When all the capacitors reach equilibrium, the charge on C_3 is found to be $5\mu C$. The value of $\in_r$. **[JEE (Advanced) 2018]**



PCC045

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7. In the circuit shown, initially there is no charge on capacitors and keys S_1 and S_2 are open. The values of the capacitors are $C_1 = 10\ \mu F$, $C_2 = 30\ \mu F$ and $C_3 = C_4 = 80\ \mu F$. **[JEE (Advanced) 2019]**



Which of the statement(s) is/are correct ?

(A) The keys S_1 is kept closed for long time such that capacitors are fully charged. Now key S_2 is closed, at this time, the instantaneous current across 30 Ω resistor (between points P and Q) will be 0.2 A (round off to 1st decimal place).

(B) If key S_1 is kept closed for long time such that capacitors are fully charged, the voltage difference between points P and Q will be 10 V.

(C) At time $t = 0$, the key S_1 is closed, the instantaneous current in the closed circuit will be 25 mA.

(D) If key S_1 is kept closed for long time such that capacitors are fully charged, the voltage across the capacitors C_1 will be 4V.

PCC046

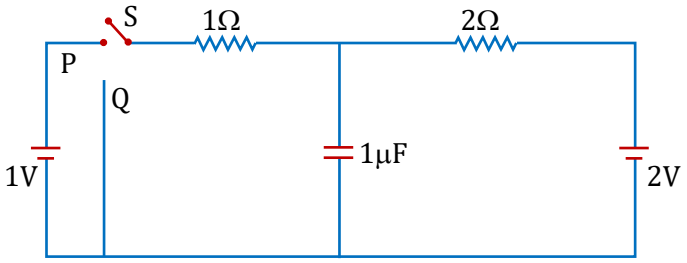
8. A parallel plate capacitor of capacitance C has spacing d between two plates having area A. The region between the plates is filled with N dielectric layers, parallel to its plates, each with thickness $\delta = \frac{d}{N}$. The dielectric constant of the mth layer is $K_m = K\left(1 + \frac{m}{N}\right)$. For A very large N (> 10^3), the capacitance C is $\alpha\left(\frac{K \in_0 A}{d\ell n 2}\right)$. The value of α will be ____. [$\in_0$ is the permittivity of free space]

[JEE (Advanced) 2019]

PCC047

QUESTION STEM FOR QUESTION NOS. 9 AND 10

In the circuit shown below, the switch S is connected to position P for a long time so that the charge on the capacitor becomes $q_1\ \mu C$. Then S is switched to position Q. After a long time, the charge on the capacitor is $q_2\ \mu C$. **[JEE (Advanced) 2021]**



9. The magnitude of q_1 is ___ .

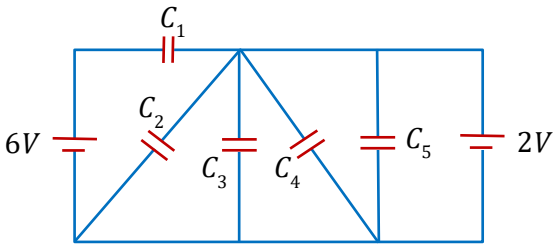
PCC048

10. The magnitude of q_2 is ___ .

PCC049

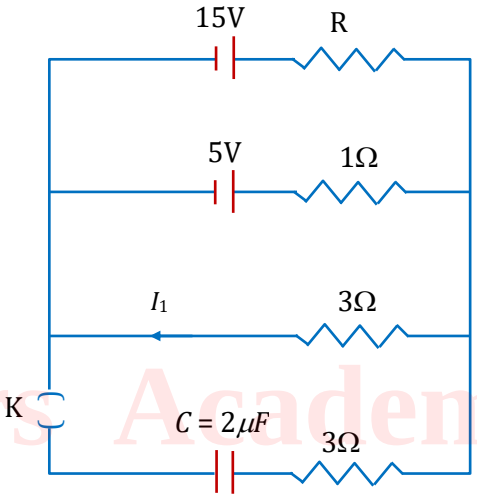
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11. In the following circuit $C_1 = 12\ \mu F$, $C_2 = C_3 = 4\ \mu F$ and $C_4 = C_5 = 2\ \mu F$. The Charge stored in C_3 is _____ μC. **[JEE (Advanced) 2022]**



PCC050

12. In a circuit shown in the figure, the capacitor C is initially uncharged and the key K is open. In this condition, a current of 1 A flows through the 1 Ω resistor. The key is closed at time $t = t_0$. Which of the following statement(s) is(are) correct?
[Given: $e^{-1} = 0.36$] **[JEE (Advanced) 2023]**

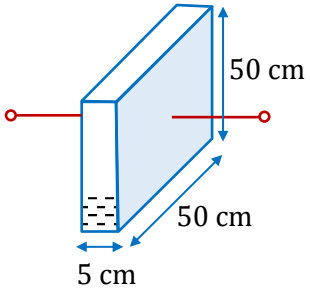


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(A) The value of the resistance R is 3Ω.
(B) For $t < t_0$, the value of current I_1 is 2A.
(C) At $t = t_0 + 7.2\ \mu s$, the current in the capacitor is 0.6 A.
(D) For $t \to \infty$, the charge on the capacitor is 12 μC.

PCC124

13. A container has a base of 50 cm × 5 cm and height 50 cm, as shown in the figure. It has two parallel electrically conducting walls each of area 50 cm × 50 cm. The remaining walls of the container are thin and non-conducting. The container is being filled with a liquid of dielectric constant 3 at a uniform rate of 250 $cm^3\ s^{-1}$. What is the value of the capacitance of the container after 10 seconds?
[Given: Permittivity of free space $\in_0 = 9 \times 10^{-12}\ C^2\ N^{-1} m^{-2}$, the effects of the non-conducting walls on the capacitance are negligible] **[JEE (Advanced) 2023]**

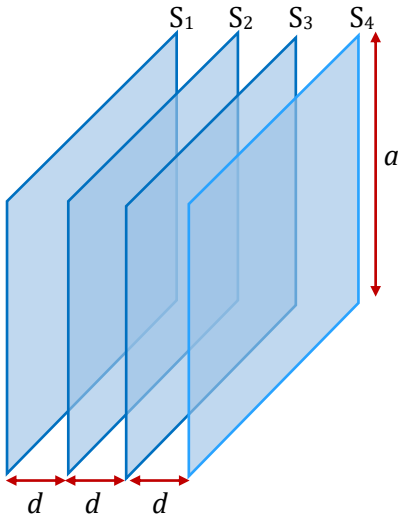


(A) 27 pF (B) 63 pF (C) 81 pF (D) 135 pF

PCC125

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14. Four identical thin, square metal sheets, S_1, S_2, S_3 and S_4, each of side a are kept parallel to each other with equal distance d (<< a) between them, as shown in the figure. Let $C_0 = \varepsilon_0 a^2/d$, where ε_0 is the permittivity of free space. **[JEE (Advanced) 2024]**



Match the quantities mentioned in List-I with their values in List-II and choose the correct option.

	List-I		**List-II**
(P)	The capacitance between S_1 and S_4, with S_2 and S_3 not connected, is	(1)	$3C_0$
(Q)	The capacitance between S_1 and S_4, with S_2 shorted to S_3, is	(2)	$C_0/2$
(R)	The capacitance between S_1 and S_3, with S_2 shorted to S_4, is	(3)	$C_0/3$
(S)	The capacitance between S_1 and S_2, with S_3 shorted to S_1, and S_2 shorted to S_4, is	(4)	$2C_0/3$
		(5)	$2C_0$

(A) P → 3; Q → 2; R → 4; S → 5

(B) P → 2; Q → 3; R → 2; S → 1

(C) P → 3; Q → 2; R → 4; S → 1

(D) P → 3; Q → 2; R → 2; S → 5

PCC149

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	C	A	D	B	B	A	A	D	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	D	B	B	A	A	B	C	B	D
Que.	21	22	23							
Ans.	A	B	C							

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	AB	BCD	ACD	BC	AB	BD	ABD	BCD	A	D
Que.	11	12	13							
Ans.	B	AC	D							

EXERCISE (S)

1. 2 **2.** 5 **3.** $12\mu C$

4. (a) $\frac{100}{7}$ volts; (b) 28.56 μC, 42.84 μC, 71.4 μC, 22.88 μC **5.** 30

6. 2.5 kV **7.** 16 **8.** 6

9. 300 **10.** 3 **11.** $\Delta q = \frac{1}{2}CE\frac{K-1}{K+1}$

12. $dC_{net} = \frac{\varepsilon_0 a k_1 k_2}{(k_1 - k_2)\frac{d}{a}} \log\left(\frac{k_1}{k_2}\right)$ **13.** $M = \frac{\varepsilon_0 b V^2 (k-1)}{2dg}$ **14.** $\frac{V_1}{V_2} = \frac{\sqrt{k_2 - 1}}{\sqrt{k_1 - 1}}$

15. $8.37W$ **16.** 18

17. $q_1 = 2\times10^{-6}C$, $q_2 = 8\times10^{-6}C$, $q_3 = 9\times10^{-6}C$, $q_4 = 12\times10^{-6}C$ **18.** 750

19. $Q_0 = \frac{CVR_2}{R_1 + R_2}$ and $\alpha = \frac{R_1 + R_2}{CR_1R_2}$ **20.** (a) $Q = \frac{CV}{2}\left(1 - e^{-2t/3RC}\right)$, (b) $i = \frac{V}{2R} - \frac{V}{6R}e^{-2t/3RC}, \frac{V}{2R}$

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	3	2	4	3	1	4	1	5	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	1	2	400	1	1	36	60	1	1320

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EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	A	ABCD	A	A	1.50	CD	0.99 to 1.01	1.33	0.67
Que.	11	12	13	14						
Ans.	8	ABCD	B	C						

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Important Notes

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